

Transnational Family Migration and Loneliness Among Older Filipinos

First-Difference + IPW Analysis with Mechanism and Exploratory Moderation

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1 Overview and Analytic Strategy

This document presents the full analysis for “**Transnational Family Migration and Loneliness Among Older Filipinos**” using the Longitudinal Study of Aging and Health in the Philippines (LSAHP), Waves 1 and 2.

1.1 Three-Tier Analytic Structure

The analysis is organised into three tiers reflecting different levels of causal identification and available sample size.

Tier 1 — Causal (H1, n = 3,291). A survey-weighted first-difference (FD) model with inverse probability weighting (IPW) estimates the within-person effect of entering a transnational household on loneliness. FD removes all time-invariant confounding; IPW additionally balances 15 observed baseline characteristics between enterers and non-enterers. The primary outcome is the continuous 3-item loneliness scale change (d_lonely3); the binary CES-D frequent loneliness item (d_lonely) serves as a robustness check.

Tier 2 — Associational / Mechanism (H2, n = 3,075). Among respondents with non-co-resident children, changes in contact satisfaction and contact frequency are added to the FD model. These estimates identify within-person associations between communication quality and loneliness change, net of transnational transition. Reverse causality (loneliness reducing satisfaction) cannot

be fully ruled out; findings are interpreted as evidence of mechanism rather than strict causal effects.

Tier 3 — Exploratory (H3 + H4, n = 395). Distance models are restricted to respondents with recorded destination countries in both waves. Crucially, this subsample consists **entirely of households that were already transnational at Wave 1 and remained so at Wave 2** — there are zero enterers in this subsample. The distance analysis therefore does not identify the entry effect; instead it asks: *among already-transnational households, does a child moving farther away increase parental loneliness, and does contact satisfaction buffer this?* This is a distinct and meaningful question, but must be clearly distinguished from the Tier 1 causal claim.

1.2 Key Missingness Facts

Variable	Missing	%
d_lonely (binary)	2,648	44%
d_lonely3 (continuous)	2,694	45%
d_contact_sat	2,869	48%
d_dist_1000km	5,513	92%
d_poor / d_urban	1,974	33%
d_health	2,692	45%

The 92% missingness on distance is structural: destination country codes are only recorded when a child is abroad at both waves with a valid country code entered by the interviewer. The distance subsample is highly selected and findings must be treated as preliminary.

2 Setup

3 Load Data

```
# Remove only data objects, not helper functions
rm(list = ls()[!ls() %in% c("LMUGreen", "theme_lmu", "save_plot")])

library(haven)

lsahp1 <- read_dta(
  "C:/Users/jemma/Downloads/LSAHP-Baseline-Data-STATA/LSAHP Baseline Data (STATA)/LSAHP1.dta"
)
```

```
lsahp2 <- read_dta(
  "H:/My Drive/UPPI_RUG/UPPI_RUG_DLSU/LSAHP W2 Transnational households and loneliness (Jem, 2018)
")

cat("Wave 1 n =", nrow(lsahp1), "\n")
```

Wave 1 n = 5985

```
cat("Wave 2 n =", nrow(lsahp2), "\n")
```

Wave 2 n = 5985

4 Wave 1 Variables

4.1 ID

```
lsahp_w1 <- lsahp1 |> dplyr::mutate(id = P1ID1)
```

4.2 Outcomes

```
# Binary: 1 = often lonely (frequent loneliness threshold)
lsahp_w1 <- lsahp_w1 |>
  dplyr::mutate(
    lonely_w1 = dplyr::case_when(
      P1SD1_06 == 3 ~ 1,
      P1SD1_06 %in% c(1, 2) ~ 0,
      TRUE ~ NA_real_
    )
  )

# 3-item continuous scale (0-4): mean of companionship, left out, isolated
# Wave 1 uses same 0-4 response categories as Wave 2 - confirmed comparable
lsahp_w1 <- lsahp_w1 |>
  dplyr::mutate(
    lonely_3item = dplyr::if_else(
      rowSums(dplyr::across(c(P1SH9_1, P1SH9_2, P1SH9_3),
        ~ is.na(.x))) == 0,
      rowMeans(dplyr::across(c(P1SH9_1, P1SH9_2, P1SH9_3))),
      NA_real_
    )
  )
```

```
)
)

cat("Wave 1 lonely_w1:\n")
```

Wave 1 lonely_w1:

```
table(lsa hp_w1$lonely_w1, useNA = "ifany")
```

```

  0    1 <NA>
4795 414  776
```

```
cat("\nWave 1 lonely_3item summary:\n")
```

Wave 1 lonely_3item summary:

```
summary(lsa hp_w1$lonely_3item)
```

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
0.0000  0.3333   1.0000  0.9453  1.3333   4.0000     775
```

4.3 Exposure: Transnational Household

```
lsa hp_w1 <- lsa hp_w1 |>
  dplyr::mutate(
    child_abroad = dplyr::if_else(
      rowSums(dplyr::across(dplyr::starts_with("P1HH39_"),
        ~ as.numeric(.x == 6)),
        na.rm = TRUE) > 0, 1L, 0L),

    # CORRECT: use P1HH42_* for child-specific OFW status
    child_ofw = dplyr::if_else(
      rowSums(dplyr::across(dplyr::starts_with("P1HH42_"),
        ~ as.numeric(.x == 1)),
        na.rm = TRUE) > 0, 1L, 0L),

    transnational_hh = dplyr::if_else(
      child_abroad == 1L | child_ofw == 1L, 1L, 0L)
  )

cat("Wave 1 transnational_hh:\n")
```

Wave 1 transnational_hh:

```
table(lsahp_w1$transnational_hh, useNA = "ifany")
```

```
  0    1  
4795 1190
```

4.4 Mediators

```
# Contact frequency: minimum J10 code across children (lower = more frequent)
lsahp_w1 <- lsahp_w1 |>
  dplyr::mutate(
    contact_child_freq = {
      j10_mat <- dplyr::across(
        dplyr::starts_with("P1SJ10_"),
        ~ dplyr::if_else(.x %in% 0:4, as.numeric(.x), NA_real_)
      ) |> as.matrix()
      apply(j10_mat, 1, function(x) {
        if (all(is.na(x))) NA_real_ else min(x, na.rm = TRUE)
      })
    }
  )

# Contact satisfaction: binary (1 = satisfied, 0 = not)
# CORRECT coding: {1,2} = satisfied, {3} = not satisfied
lsahp_w1 <- lsahp_w1 |>
  dplyr::mutate(
    contact_child_satisfied = dplyr::case_when(
      P1SJ26 %in% c(1, 2) ~ 1L,
      P1SJ26 == 3 ~ 0L,
      TRUE ~ NA_integer_
    )
  )
```

4.5 Distance to Farthest Migrant Child

```
library(rnaturalearth); library(sf); library(geosphere)
sf::sf_use_s2(FALSE)

ref_lon <- 120.9842; ref_lat <- 14.5995

world <- rnaturalearth::ne_countries(scale = "medium", returnclass = "sf")
```

```

world_centroids <- world |>
  dplyr::mutate(centroid = sf::st_centroid(geometry)) |>
  dplyr::mutate(lon = sf::st_coordinates(centroid)[,"X"],
               lat = sf::st_coordinates(centroid)[,"Y"]) |>
  dplyr::select(iso_a2, name, lon, lat)

country_vars_w1 <- lsahp_w1 |>
  dplyr::select(dplyr::starts_with("P1HH43_")) |> names()

country_codes_w1 <- lsahp_w1 |>
  dplyr::select(dplyr::all_of(country_vars_w1)) |>
  tidyr::pivot_longer(cols = dplyr::everything(), values_to = "country_code") |>
  dplyr::filter(!is.na(country_code), country_code != "") |>
  dplyr::distinct(country_code) |>
  dplyr::left_join(world_centroids, by = c("country_code" = "iso_a2"))

lsahp_w1 <- lsahp_w1 |>
  dplyr::rowwise() |>
  dplyr::mutate(
    max_child_distance_km = {
      codes <- c_across(dplyr::all_of(country_vars_w1))
      codes <- codes[!is.na(codes) & codes != ""]
      if (length(codes) == 0) { NA_real_ } else {
        coords <- country_codes_w1 |>
          dplyr::filter(country_code %in% codes, !is.na(lon), !is.na(lat))
        if (nrow(coords) == 0) { NA_real_ } else {
          max(geosphere::distHaversine(
            matrix(c(ref_lon, ref_lat), ncol = 2),
            as.matrix(coords[, c("lon", "lat")])
          )) / 1000
        }
      }
    }
  ) |>
  dplyr::ungroup()

```

4.6 Controls and IPW Covariates

```

lsahp_w1 <- lsahp_w1 |>
  dplyr::mutate(
    urban      = dplyr::case_when(P1MX_URBAN == 1 ~ 1L,
                                   P1MX_URBAN == 2 ~ 0L, TRUE ~ NA_integer_),
    age        = as.numeric(P1SA2),
    sex        = dplyr::case_when(P1SA3 == 1 ~ 0L, P1SA3 == 2 ~ 1L,
                                   TRUE ~ NA_integer_),

```

```

educhsplus = dplyr::case_when(P1SQ16_EDUC %in% c(2,3) ~ 1L,
                              P1SQ16_EDUC == 1 ~ 0L, TRUE ~ NA_integer_),
wealthq     = as.integer(P1MX_WINDEX),
poor        = dplyr::case_when(P1MX_WINDEX == 1 ~ 1L,
                              P1MX_WINDEX %in% 2:5 ~ 0L, TRUE ~ NA_integer_),
health_3cat = dplyr::case_when(P1SB1 %in% c(1,2) ~ 0L, P1SB1 == 3 ~ 1L,
                              P1SB1 %in% c(4,5) ~ 2L, TRUE ~ NA_integer_),
marital     = dplyr::case_when(P1SA27 == 2 ~ 1L,
                              P1SA27 %in% c(1,3,4,5) ~ 0L, TRUE ~ NA_integer_),
n_children  = as.numeric(P1HH30),
remittance  = dplyr::case_when(P1SF2_08 == 1 ~ 1L, P1SF2_08 == 2 ~ 0L,
                              TRUE ~ NA_integer_),
internet    = dplyr::case_when(P1SH16 == 1 ~ 1L, P1SH16 == 2 ~ 0L,
                              TRUE ~ NA_integer_),
cellphone   = dplyr::case_when(P1SH20 == 1 ~ 1L, P1SH20 == 2 ~ 0L,
                              TRUE ~ NA_integer_),
cognitive   = as.numeric(P1SL_TOTAL),
social_org  = dplyr::case_when(P1SH4 == 1 | P1SH6 == 1 ~ 1L, TRUE ~ 0L)
)

```

5 Wave 2 Variables

```

lsahp_w2 <- lsahp2 |>
  dplyr::mutate(id = P1ID1)

# Binary loneliness
lsahp_w2 <- lsahp_w2 |>
  dplyr::mutate(
    lonely_w2 = dplyr::case_when(
      P2SD1_06 == 3 ~ 1L,
      P2SD1_06 %in% c(1, 2) ~ 0L,
      TRUE ~ NA_integer_
    )
  )

# 3-item scale
# Wave 2 uses same 0-4 scale as Wave 1 (confirmed from codebook)
# NA = 2,567 at Wave 2 vs 775 at Wave 1 - attrition driven, not scale change
lsahp_w2 <- lsahp_w2 |>
  dplyr::mutate(
    lonely_3item_w2 = dplyr::if_else(
      rowSums(dplyr::across(c(P2SH9_1, P2SH9_2, P2SH9_3),

```



```

        ~ is.na(.x))) == 0,
    rowMeans(dplyr::across(c(P2SH9_1, P2SH9_2, P2SH9_3))),
    NA_real_
  )
)

# Transnational exposure
lsahp_w2 <- lsahp_w2 |>
dplyr::mutate(
  child_abroad_w2 = dplyr::if_else(
    rowSums(dplyr::across(dplyr::starts_with("P2HH39_"),
      ~ as.numeric(.x == 6))), na.rm = TRUE) > 0, 1L, 0L),

  # CORRECTED: P2HH42_* not P2HH14_* for child-specific OFW
  child_ofw_w2 = dplyr::if_else(
    rowSums(dplyr::across(dplyr::starts_with("P2HH42_"),
      ~ as.numeric(.x == 1))), na.rm = TRUE) > 0, 1L, 0L),

  transnational_hh_w2 = dplyr::if_else(
    child_abroad_w2 == 1L | child_ofw_w2 == 1L, 1L, 0L)
)

# Contact frequency
lsahp_w2 <- lsahp_w2 |>
dplyr::mutate(
  contact_child_freq_w2 = {
    j10_mat <- dplyr::across(
      dplyr::starts_with("P2SJ10"),
      ~ dplyr::if_else(.x %in% 0:4, as.numeric(.x), NA_real_)
    ) |> as.matrix()
    apply(j10_mat, 1, function(x) {
      if (all(is.na(x))) NA_real_ else min(x, na.rm = TRUE)
    })
  }
)

# Contact satisfaction
# CORRECTED: {1,2} = satisfied, {3} = not satisfied (same as Wave 1)
lsahp_w2 <- lsahp_w2 |>
dplyr::mutate(
  contact_child_satisfied_w2 = dplyr::case_when(
    P2SJ26 %in% c(1, 2) ~ 1L,
    P2SJ26 == 3 ~ 0L,
    TRUE ~ NA_integer_
  )
)

```

```

# Controls
lsahp_w2 <- lsahp_w2 |>
  dplyr::mutate(
    urban_w2 = dplyr::case_when(P2MX_URBAN == 1 ~ 1L,
                                P2MX_URBAN == 0 ~ 0L, TRUE ~ NA_integer_),
    age_w2 = as.numeric(P2SA2),
    sex_w2 = dplyr::case_when(P2SA3 == 1 ~ 0L, P2SA3 == 2 ~ 1L,
                              TRUE ~ NA_integer_),
    educhsplus_w2 = dplyr::case_when(w2op_educ3 %in% c(2,3) ~ 1L,
                                      w2op_educ3 == 1 ~ 0L, TRUE ~ NA_integer_),
    wealthq_w2 = as.integer(P2MX_WINDEX_A),
    poor_w2 = dplyr::case_when(P2MX_WINDEX_A == 1 ~ 1L,
                               P2MX_WINDEX_A %in% 2:5 ~ 0L, TRUE ~ NA_integer_),
    health_3cat_w2 = dplyr::case_when(P2SB1 %in% c(1,2) ~ 0L,
                                       P2SB1 == 3 ~ 1L,
                                       P2SB1 %in% c(4,5) ~ 2L, TRUE ~ NA_integer_)
  )

```

5.1 Wave 2 Distance

```

sf::sf_use_s2(FALSE)

world_centroids_n3 <- world |>
  dplyr::mutate(centroid = sf::st_centroid(geometry)) |>
  dplyr::mutate(lon = sf::st_coordinates(centroid)[,"X"],
               lat = sf::st_coordinates(centroid)[,"Y"]) |>
  dplyr::select(iso_n3, name, lon, lat) |>
  dplyr::mutate(iso_n3 = as.numeric(iso_n3))

country_vars_w2 <- lsahp_w2 |>
  dplyr::select(dplyr::starts_with("P2HH43_")) |> names()

country_codes_w2 <- lsahp_w2 |>
  dplyr::select(dplyr::all_of(country_vars_w2)) |>
  tidyr::pivot_longer(cols = dplyr::everything(), values_to = "country_code") |>
  dplyr::mutate(country_code = as.numeric(country_code)) |>
  dplyr::filter(!is.na(country_code), country_code > 0) |>
  dplyr::distinct(country_code) |>
  dplyr::left_join(world_centroids_n3, by = c("country_code" = "iso_n3"))

lsahp_w2 <- lsahp_w2 |>
  dplyr::rowwise() |>
  dplyr::mutate(
    max_child_distance_km_w2 = {
      codes <- as.numeric(c_across(dplyr::all_of(country_vars_w2)))
    }
  )

```

```

codes <- codes[!is.na(codes) & codes > 0]
if (length(codes) == 0) { NA_real_ } else {
  coords <- country_codes_w2 |>
    dplyr::filter(country_code %in% codes, !is.na(lon), !is.na(lat))
  if (nrow(coords) == 0) { NA_real_ } else {
    max(geosphere::distHaversine(
      matrix(c(ref_lon, ref_lat), ncol = 2),
      as.matrix(coords[, c("lon", "lat")])
    )) / 1000
  }
}
}
) |>
dplyr::ungroup()

```

6 Panel Construction

```

panel_ids <- intersect(lsahp_w1$id, lsahp_w2$id)
cat("Balanced panel n =", length(panel_ids), "\n")

```

Balanced panel n = 5985

```

lsahp_w1p <- lsahp_w1 |> dplyr::filter(id %in% panel_ids)
lsahp_w2p <- lsahp_w2 |> dplyr::filter(id %in% panel_ids)

# Long format
lsahp_w1_long <- lsahp_w1p |>
  dplyr::transmute(
    id, wave = 1L,
    lonely = as.numeric(lonely_w1), lonely_3item,
    transnational_hh, contact_child_freq, contact_child_satisfied,
    max_child_distance_km, urban, age, sex, educhsplus,
    wealthq, poor, health_3cat,
    strata = P1MX_STRATA, psu = P1MX_PSU, wgt = P1MX_WGTU
  )

lsahp_w2_long <- lsahp_w2p |>
  dplyr::transmute(
    id, wave = 2L,
    lonely = as.numeric(lonely_w2), lonely_3item = lonely_3item_w2,
    transnational_hh = transnational_hh_w2,

```

```

    contact_child_freq = contact_child_freq_w2,
    contact_child_satisfied = contact_child_satisfied_w2,
    max_child_distance_km = max_child_distance_km_w2,
    urban = urban_w2, age = age_w2, sex = sex_w2,
    educhsplus = educhsplus_w2, wealthq = wealthq_w2,
    poor = poor_w2, health_3cat = health_3cat_w2,
    strata = P1MX_STRATA, psu = P1MX_PSU, wgt = P2MX_WGT1
  )

lsahp_long <- dplyr::bind_rows(lsahp_w1_long, lsahp_w2_long)

# Wide format
lsahp_wide <- lsahp_long |>
  dplyr::select(id, wave, lonely, lonely_3item, transnational_hh,
               contact_child_freq, contact_child_satisfied,
               max_child_distance_km) |>
  tidyr::pivot_wider(names_from = wave,
                    values_from = c(lonely, lonely_3item, transnational_hh,
                                    contact_child_freq, contact_child_satisfied,
                                    max_child_distance_km),
                    names_sep = "_w")

# Controls wide
controls_wide <- lsahp_long |>
  dplyr::select(id, wave, poor, urban, health_3cat) |>
  tidyr::pivot_wider(names_from = wave,
                    values_from = c(poor, urban, health_3cat),
                    names_sep = "_w")

lsahp_wide <- lsahp_wide |>
  dplyr::left_join(controls_wide, by = "id")

# Transition indicators and first differences
lsahp_wide <- lsahp_wide |>
  dplyr::mutate(
    trans_transition = dplyr::case_when(
      transnational_hh_w1 == 0 & transnational_hh_w2 == 0 ~ "Stayed non-transnational",
      transnational_hh_w1 == 0 & transnational_hh_w2 == 1 ~ "Entered transnational",
      transnational_hh_w1 == 1 & transnational_hh_w2 == 1 ~ "Stayed transnational",
      transnational_hh_w1 == 1 & transnational_hh_w2 == 0 ~ "Exited transnational",
      TRUE ~ NA_character_
    ),
    entered = dplyr::if_else(
      transnational_hh_w1 == 0 & transnational_hh_w2 == 1, 1L, 0L),
    exited = dplyr::if_else(
      transnational_hh_w1 == 1 & transnational_hh_w2 == 0, 1L, 0L),
    # First differences

```

```

d_lonely      = as.integer(lonely_w2) - as.integer(lonely_w1),
d_lonely3     = lonely_3item_w2 - lonely_3item_w1,
d_contact_sat = contact_child_satisfied_w2 - contact_child_satisfied_w1,
d_contact_freq = contact_child_freq_w2 - contact_child_freq_w1,
d_dist_km     = max_child_distance_km_w2 - max_child_distance_km_w1,
d_dist_1000km = d_dist_km / 1000,
d_poor        = poor_w2 - poor_w1,
d_urban       = urban_w2 - urban_w1,
d_health      = health_3cat_w2 - health_3cat_w1,
# Interaction term (satisfaction × distance)
d_sat_x_dist  = d_contact_sat * d_dist_1000km,
# Became lonely
became_lonely = dplyr::if_else(d_lonely == 1, 1L, 0L)
)

cat("Wide n =", nrow(lsahp_wide), "\n")

```

Wide n = 5985

```
cat("\nTransition distribution:\n")
```

Transition distribution:

```
table(lsahp_wide$trans_transition, useNA = "ifany")
```

Entered transnational	Exited transnational	Stayed non-transnational
254	644	4541
Stayed transnational		
546		

7 Missingness Audit

```

lsahp_wide |>
dplyr::summarise(
  n_total      = dplyr::n(),
  miss_d_lonely = sum(is.na(d_lonely)),
  miss_d_lonely3 = sum(is.na(d_lonely3)),
  miss_d_contact_sat = sum(is.na(d_contact_sat)),

```

```

miss_d_dist_1000km = sum(is.na(d_dist_1000km)),
miss_d_poor        = sum(is.na(d_poor)),
miss_d_urban       = sum(is.na(d_urban)),
miss_d_health      = sum(is.na(d_health)),
n_tier1_cont       = sum(!is.na(d_lonely3) & !is.na(entered) &
                        !is.na(d_poor) & !is.na(d_urban) &
                        !is.na(d_health)),
n_tier2_contact    = sum(!is.na(d_lonely3) & !is.na(entered) &
                        !is.na(d_contact_sat) &
                        !is.na(d_poor) & !is.na(d_urban) &
                        !is.na(d_health)),
n_tier3_distance   = sum(!is.na(d_lonely3) & !is.na(entered) &
                        !is.na(d_dist_1000km) &
                        !is.na(d_poor) & !is.na(d_urban) &
                        !is.na(d_health)),
n_tier3_interaction = sum(!is.na(d_lonely3) & !is.na(entered) &
                        !is.na(d_contact_sat) & !is.na(d_dist_1000km) &
                        !is.na(d_poor) & !is.na(d_urban) &
                        !is.na(d_health))
) |>
tidyr::pivot_longer(dplyr::everything()) |>
knitr::kable(caption = "Missingness audit and analytic sample sizes by tier")

```

Table 2: Missingness audit and analytic sample sizes by tier

name	value
n_total	5985
miss_d_lonely	2648
miss_d_lonely3	2649
miss_d_contact_sat	2869
miss_d_dist_1000km	5513
miss_d_poor	1974
miss_d_urban	1974
miss_d_health	2692
n_tier1_cont	3291
n_tier2_contact	3075
n_tier3_distance	395
n_tier3_interaction	393

```

# CRITICAL: confirm zero enterers in distance subsample
cat("\n=== Enterer breakdown in distance subsample ===\n")

```

```

=== Enterer breakdown in distance subsample ===

```

```
lsahp_wide |>
  dplyr::filter(!is.na(d_lonely3) & !is.na(d_dist_1000km) &
    !is.na(d_poor) & !is.na(d_urban) & !is.na(d_health)) |>
  dplyr::count(entered, exited) |>
  dplyr::mutate(group = dplyr::case_when(
    entered == 1 ~ "Enterers (0→1)",
    exited == 1 ~ "Exiters (1→0)",
    TRUE ~ "Stayers (no change)"
  )) |>
  knitr::kable(caption = "Transition groups in distance subsample (n=395). NOTE: zero enterers
```

Table 3: Transition groups in distance subsample (n=395). NOTE: zero enterers — distance analysis is within already-transnational households only.

entered	exited	n	group
0	0	395	Stayers (no change)

8 Descriptives

8.1 Survey Designs

```
library(survey)

des_w1 <- svydesign(ids = ~psu, strata = ~strata, weights = ~wgt,
  data = dplyr::filter(lsahp_long, wave == 1, !is.na(wgt)),
  nest = TRUE)

des_w2 <- svydesign(ids = ~psu, strata = ~strata, weights = ~wgt,
  data = dplyr::filter(lsahp_long, wave == 2, !is.na(wgt)),
  nest = TRUE)

design_w1_vars <- lsahp_long |>
  dplyr::filter(wave == 1) |>
  dplyr::select(id, strata, psu, wgt) |>
  dplyr::distinct()

lsahp_wide_des <- lsahp_wide |>
  dplyr::left_join(design_w1_vars, by = "id")

des_wide <- svydesign(ids = ~psu, strata = ~strata, weights = ~wgt,
  data = lsahp_wide_des, nest = TRUE)
```

```
svymean_ci <- function(var, design, level = 0.95) {
  m <- survey::svymean(var, design, na.rm = TRUE)
  ci <- suppressMessages(confint(m, level = level))
  data.frame(mean = as.numeric(coef(m)),
             ci_l = as.numeric(ci[,1]),
             ci_u = as.numeric(ci[,2]))
}
```

8.2 Figure 1: Loneliness Prevalence

```
prev_lon <- dplyr::bind_rows(
  svymean_ci(~lonely, des_w1) |> dplyr::mutate(Wave = "Wave 1"),
  svymean_ci(~lonely, des_w2) |> dplyr::mutate(Wave = "Wave 2")
) |> dplyr::mutate(label = scales::percent(mean, accuracy = 0.1))

p_lon <- ggplot(prev_lon, aes(x = Wave, y = mean)) +
  geom_col(fill = LMUGreen, color = "#006B2C", width = 0.6) +
  geom_errorbar(aes(ymin = ci_l, ymax = ci_u), width = 0.12, linewidth = 0.9) +
  geom_text(aes(y = ci_u + 0.015, label = label), size = 6, fontface = "bold") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1),
                     limits = c(0, 0.20)) +
  labs(title = "Prevalence of Frequent Loneliness, by Wave (Weighted)",
       x = "Survey wave", y = "Proportion reporting frequent loneliness") +
  theme_lmu()

save_plot(p_lon, "01_lonely_prevalence")
p_lon
```

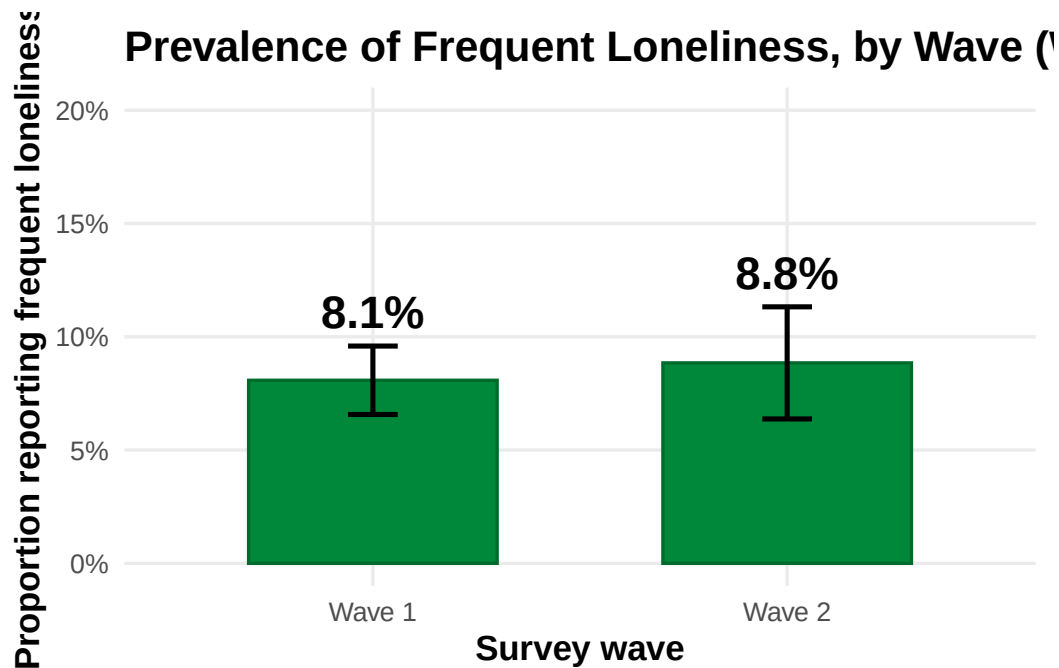



Figure 1: Weighted prevalence of frequent loneliness (often = 1 on CES-D) by survey wave.

8.3 Figure 2: Transnational Prevalence

```
prev_tn <- dplyr::bind_rows(
  svymean_ci(~transnational_hh, des_w1) |> dplyr::mutate(Wave = "Wave 1"),
  svymean_ci(~transnational_hh, des_w2) |> dplyr::mutate(Wave = "Wave 2")
) |> dplyr::mutate(label = scales::percent(mean, accuracy = 0.1))

p_tn <- ggplot(prev_tn, aes(x = Wave, y = mean)) +
  geom_col(fill = LMUGreen, color = "#006B2C", width = 0.6) +
  geom_errorbar(aes(ymin = ci_l, ymax = ci_u), width = 0.12, linewidth = 0.9) +
  geom_text(aes(y = ci_u + 0.03, label = label), size = 6, fontface = "bold") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1),
    limits = c(0, 0.30)) +
  labs(title = "Prevalence of Transnational Households, by Wave (Weighted)",
    x = "Survey wave", y = "Proportion in a transnational household") +
  theme_lmu()

save_plot(p_tn, "02_transnational_prevalence")
p_tn
```

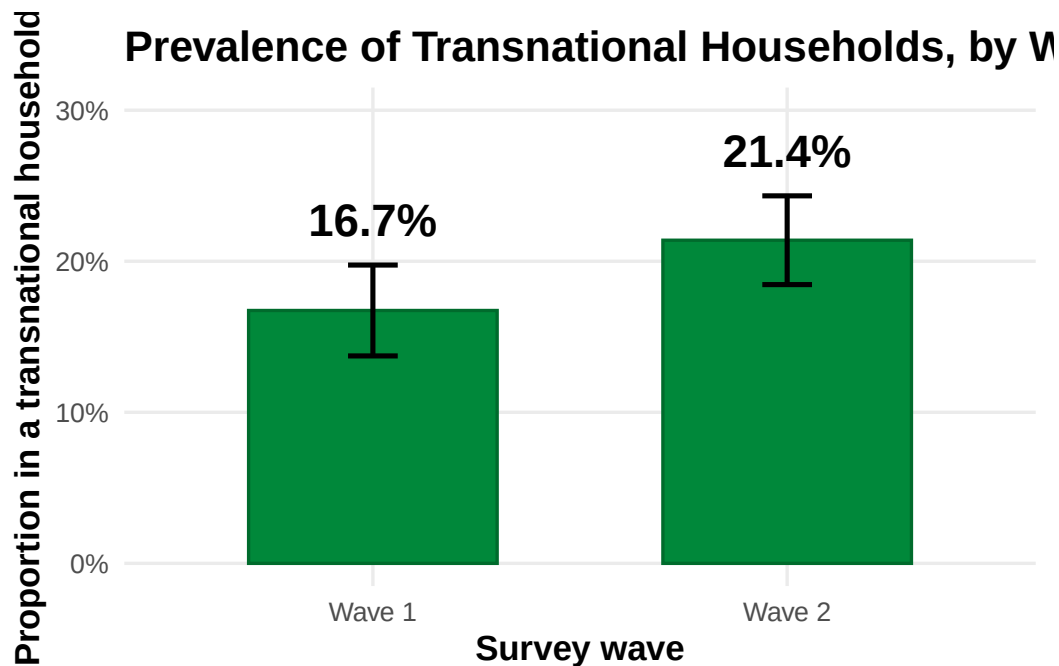


Figure 2: Weighted prevalence of transnational households by survey wave.

8.4 Figure 3: Transition Distribution

```
m_trans <- svymean(~factor(trans_transition), des_wide, na.rm = TRUE)
ci_trans <- suppressMessages(confint(m_trans))

trans_df <- data.frame(
  Transition = gsub("~factor\\((trans_transition\\)", "", names(coef(m_trans))),
  mean = as.numeric(coef(m_trans)),
  ci_l = as.numeric(ci_trans[,1]),
  ci_u = as.numeric(ci_trans[,2])
) |> dplyr::mutate(label = scales::percent(mean, accuracy = 0.1))

p_trans <- ggplot(trans_df, aes(x = reorder(Transition, -mean), y = mean)) +
  geom_col(fill = LMUGreen, color = "#006B2C", width = 0.7) +
  geom_errorbar(aes(ymin = ci_l, ymax = ci_u), width = 0.12, linewidth = 0.9) +
  geom_text(aes(y = ci_u + 0.03, label = label), size = 6, fontface = "bold") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1),
    limits = c(0, 0.90)) +
  labs(title = "Transnational Household Transitions (Weighted)",
    x = "Transition group (Wave 1 → Wave 2)",
    y = "Proportion of the sample") +
  theme_lmu() +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```

```
save_plot(p_trans, "03_transitions", w = 10, h = 6)
p_trans
```

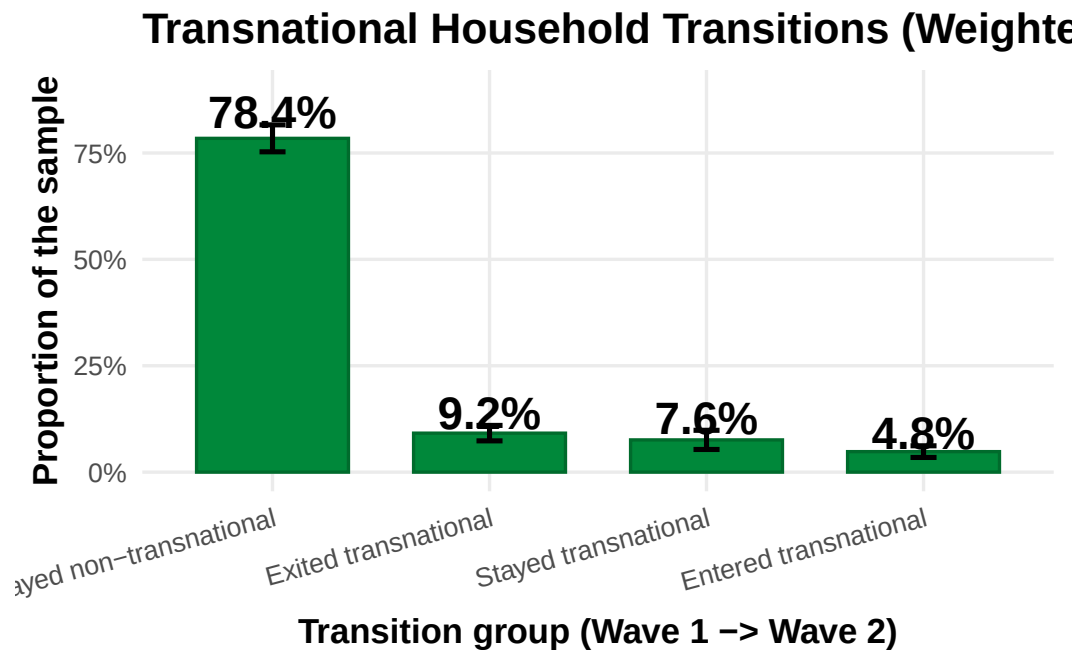


Figure 3: Weighted distribution of household transitions between Wave 1 and Wave 2.

8.5 Figure 4: Becoming Lonely by Transition

```
svy_mean_by_group <- function(design, yvar, gvar) {
  levs <- levels(factor(design$variables[[gvar]]))
  dplyr::bind_rows(lapply(levs, function(L) {
    dsub <- subset(design, design$variables[[gvar]] == L)
    m <- svymean(as.formula(paste0("~", yvar)), dsub, na.rm = TRUE)
    ci <- suppressMessages(confint(m))
    data.frame(group = L, mean = as.numeric(coef(m)),
               ci_l = as.numeric(ci[,1]), ci_u = as.numeric(ci[,2]))
  }))
}

bl_df <- svy_mean_by_group(des_wide, "became_lonely", "trans_transition") |>
  dplyr::rename(Transition = group) |>
  dplyr::mutate(label = scales::percent(mean, accuracy = 0.1))

p_bl <- ggplot(bl_df, aes(x = reorder(Transition, -mean), y = mean)) +
  geom_col(fill = LMUGreen, color = "#006B2C", width = 0.7) +
  geom_errorbar(aes(ymin = ci_l, ymax = ci_u), width = 0.12, linewidth = 0.9) +
```

```

geom_text(aes(y = ci_u + 0.04, label = label), size = 6, fontface = "bold") +
scale_y_continuous(labels = scales::percent_format(accuracy = 1),
                    limits = c(0, 0.45)) +
labs(title = "Percent Becoming Lonely by Transnational Transition (Weighted)",
     x = "Transition group (Wave 1 → Wave 2)",
     y = "Proportion who became frequently lonely") +
theme_lmu() +
theme(axis.text.x = element_text(angle = 15, hjust = 1))

save_plot(p_bl, "04_became_lonely", w = 10, h = 6)
p_bl

```

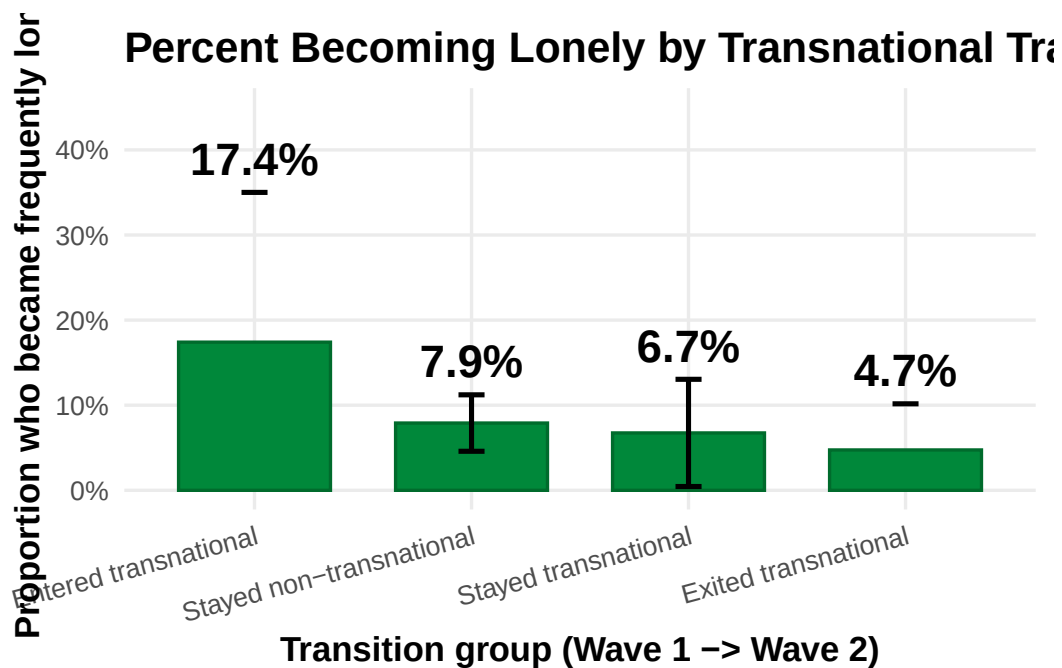


Figure 4: Weighted proportion becoming frequently lonely by transition group.

9 TIER 1: Causal Analysis (H1)

9.1 IPW — Propensity Score Estimation

```

library(WeightIt); library(cobalt)

baseline_vars <- lsahp_w1 |>
  dplyr::transmute(

```

```

    id,
    age_w1      = age,
    sex_w1      = sex,
    educ_w1     = educhsplus,
    wealth_w1   = wealthq,
    marital_w1  = marital,
    health_w1   = health_3cat,
    n_children_w1 = n_children,
    # remittance_w1 REMOVED - 97% missing, not usable
    internet_w1 = internet,
    cellphone_w1 = cellphone,
    cognitive_w1 = cognitive,
    lonely3_w1  = lonely_3item,
    social_org_w1 = social_org
    # urban_w1 and poor_w1 already in lsahp_wide_des - not joined here
  )

lsahp_wide_des2 <- lsahp_wide_des |>
  dplyr::left_join(baseline_vars, by = "id")

# IPW sample: baseline non-transnational only
lsahp_ipw <- lsahp_wide_des2 |>
  dplyr::filter(transnational_hh_w1 == 0) |>
  dplyr::mutate(
    d_dist_1000km = d_dist_km / 1000,
    d_sat_x_dist  = d_contact_sat * d_dist_1000km
  )

# Mean imputation for PS covariates with <20% missingness
# Standard practice for propensity score estimation
# Reported in methods: "Missing values on PS covariates imputed
# to the covariate mean prior to propensity score estimation"
lsahp_ipw <- lsahp_ipw |>
  dplyr::mutate(
    educ_w1      = dplyr::if_else(
      is.na(educ_w1),
      mean(educ_w1, na.rm = TRUE), as.numeric(educ_w1)),
    health_w1    = dplyr::if_else(
      is.na(health_w1),
      mean(health_w1, na.rm = TRUE), as.numeric(health_w1)),
    n_children_w1 = dplyr::if_else(
      is.na(n_children_w1),
      mean(n_children_w1, na.rm = TRUE), as.numeric(n_children_w1)),
    cognitive_w1  = dplyr::if_else(
      is.na(cognitive_w1),
      mean(cognitive_w1, na.rm = TRUE), as.numeric(cognitive_w1)),
    lonely3_w1    = dplyr::if_else(

```

```

    is.na(lonely3_w1),
    mean(lonely3_w1, na.rm = TRUE), as.numeric(lonely3_w1))
)

cat("IPW sample: n =", nrow(lsahp_ipw),
    "| Enterers:", sum(lsahp_ipw$entered, na.rm = TRUE),
    "| Stayers:", sum(lsahp_ipw$entered == 0, na.rm = TRUE), "\n")

```

IPW sample: n = 4795 | Enterers: 254 | Stayers: 4541

```

# Confirm no remaining missingness on PS covariates
ps_vars <- c("age_w1", "sex_w1", "educ_w1", "wealth_w1", "urban_w1",
             "marital_w1", "health_w1", "n_children_w1",
             "internet_w1", "cellphone_w1",
             "cognitive_w1", "lonely3_w1", "social_org_w1",
             "poor_w1")

remaining_miss <- lsahp_ipw |>
  dplyr::select(dplyr::all_of(ps_vars)) |>
  dplyr::summarise(dplyr::across(dplyr::everything(), ~ sum(is.na(.x)))) |>
  tidyr::pivot_longer(dplyr::everything()) |>
  dplyr::filter(value > 0)

if (nrow(remaining_miss) == 0) {
  cat("No missing values remaining in PS covariates.\n")
} else {
  print(remaining_miss)
}

```

No missing values remaining in PS covariates.

```

# Updated formula: remittance_w1 removed (97% missing)
ps_formula <- entered ~
  age_w1 + sex_w1 + educ_w1 + wealth_w1 + urban_w1 +
  marital_w1 + health_w1 + n_children_w1 +
  poor_w1 + internet_w1 + cellphone_w1 +
  cognitive_w1 + lonely3_w1 + social_org_w1

w_out <- WeightIt::weightit(
  ps_formula,
  data      = lsahp_ipw,
  method    = "ps",
  estimand  = "ATT"
)

summary(w_out)

```

Summary of weights

- Weight ranges:

	Min	Max
treated	1.	1.
control	0.008	0.44

- Units with the 5 most extreme weights by group:

	1	2	3	4	5
treated	1	1	1	1	1
	4024	3786	1094	523	413
control	0.287	0.298	0.345	0.357	0.44

- Weight statistics:

	Coef of Var	MAD	Entropy	# Zeros
treated	0.000	0.000	0.000	0
control	0.687	0.496	0.195	0

- Effective Sample Sizes:

	Control	Treated
Unweighted	4541.	254
Weighted	3085.98	254

```
# Combine IPW with survey weights
lsahp_ipw <- lsahp_ipw |>
  dplyr::mutate(
    ipw = w_out$weights,
    # Check if trimming needed
    p99 = quantile(w_out$weights, 0.99, na.rm = TRUE),
    ipw_trimmed = pmin(ipw, p99),
    wgt_combined = ipw_trimmed * wgt
  )

cat("\nIPW weight summary:\n")
```

IPW weight summary:

```
summary(lsahp_ipw$ipw)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.008147	0.030687	0.047145	0.105937	0.077153	1.000000

```
cat("\n99th percentile:", round(quantile(lsaipw$ipw, 0.99), 3),
    "- trimming",
    ifelse(quantile(lsaipw$ipw, 0.99) > 5, "RECOMMENDED", "not needed"), "\n")
```

99th percentile: 1 - trimming not needed

9.2 Balance Assessment

```
bal_tab <- cobalt::bal.tab(
  w_out,
  stats      = c("mean.diffs", "variance.ratios"),
  thresholds = c(m = 0.1)
)
print(bal_tab)
```

Balance Measures

	Type	Diff.Adj	M.Threshold	V.Ratio.Adj
prop.score	Distance	0.0025	Balanced, <0.1	1.0869
age_w1	Contin.	-0.0006	Balanced, <0.1	0.9460
sex_w1	Binary	0.0000	Balanced, <0.1	.
educ_w1	Contin.	0.0019	Balanced, <0.1	1.0487
wealth_w1	Contin.	-0.0027	Balanced, <0.1	0.9589
urban_w1	Binary	-0.0017	Balanced, <0.1	.
marital_w1	Binary	0.0008	Balanced, <0.1	.
health_w1	Contin.	-0.0012	Balanced, <0.1	0.9993
n_children_w1	Contin.	0.0009	Balanced, <0.1	0.8937
poor_w1	Binary	0.0001	Balanced, <0.1	.
internet_w1	Binary	-0.0012	Balanced, <0.1	.
cellphone_w1	Binary	0.0017	Balanced, <0.1	.
cognitive_w1	Contin.	0.0012	Balanced, <0.1	0.9947
lonely3_w1	Contin.	-0.0040	Balanced, <0.1	1.0278
social_org_w1	Binary	0.0010	Balanced, <0.1	.

Balance tally for mean differences

	count
Balanced, <0.1	15
Not Balanced, >0.1	0

Variable with the greatest mean difference

Variable	Diff.Adj	M.Threshold
lonely3_w1	-0.004	Balanced, <0.1

Effective sample sizes

	Control	Treated
Unadjusted	4541.	254
Adjusted	3085.98	254

```
cobalt::love.plot(
  w_out,
  threshold = 0.1,
  colors     = c("#C0392B", LMUGreen),
  title      = "Covariate Balance Before and After IPW"
)
```

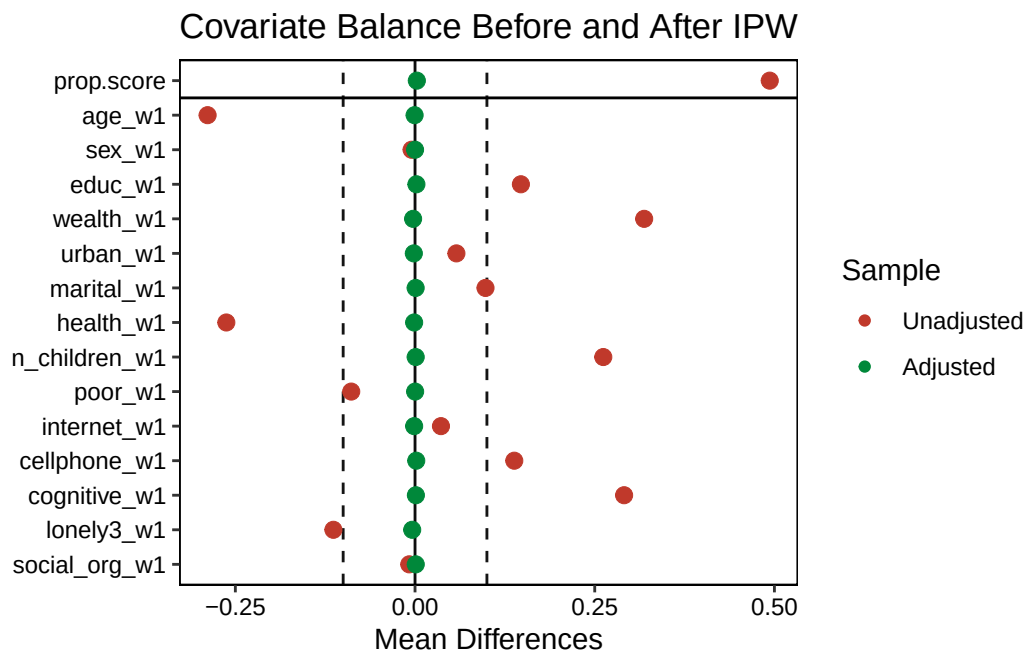


Figure 5: Covariate balance before and after IPW. All adjusted SMDs should fall within the dashed lines at ± 0.10 .

9.3 Trim and Combine Weights

```
p99 <- quantile(w_out$weights, 0.99, na.rm = TRUE)

lsahp_ipw <- lsahp_ipw |>
  dplyr::mutate(
    ipw          = w_out$weights,
    ipw_trimmed  = pmin(ipw, p99),
    wgt_combined = ipw_trimmed * wgt
  )

cat("Weight summary after trimming at 99th percentile:\n")
```

Weight summary after trimming at 99th percentile:

```
summary(lsahp_ipw$ipw_trimmed)
```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.008147	0.030687	0.047145	0.105937	0.077153	1.000000

```
# Correct ESS reporting
ess_treated <- sum(lsahp_ipw$ipw[lsahp_ipw$entered == 1])^2 /
               sum(lsahp_ipw$ipw[lsahp_ipw$entered == 1]^2)

ess_controls <- sum(lsahp_ipw$ipw[lsahp_ipw$entered == 0])^2 /
               sum(lsahp_ipw$ipw[lsahp_ipw$entered == 0]^2)

cat("Effective sample size:\n")
```

Effective sample size:

```
cat("  Treated (enterers):", round(ess_treated, 0), "\n")
```

Treated (enterers): 254

```
cat("  Controls (stayers):", round(ess_controls, 0), "\n")
```

Controls (stayers): 3086

```
cat("  Total:                ", round(ess_treated + ess_controls, 0), "\n")
```

Total: 3340

```
cat("\nControl weight range:",
    round(min(lsahp_ipw$ipw[lsahp_ipw$entered == 0]), 3), "to",
    round(max(lsahp_ipw$ipw[lsahp_ipw$entered == 0]), 3), "\n")
```

Control weight range: 0.008 to 0.44

```
cat("Trimming required: NO (max weight = 0.440)\n")
```

Trimming required: NO (max weight = 0.440)

9.4 Tier 1 Models

```

# Plain FD (comparison baseline) - IPW sample only
des_plain <- svydesign(ids = ~psu, strata = ~strata, weights = ~wgt,
                     data = lsahp_ipw, nest = TRUE)

# IPW-weighted FD (main causal estimate)
des_ipw <- svydesign(ids = ~psu, strata = ~strata, weights = ~wgt_combined,
                   data = lsahp_ipw, nest = TRUE)

# PRIMARY OUTCOME: 3-item continuous scale
# NOTE: exited removed - IPW sample is baseline non-transnational only
#       so exited = 0 for everyone (no variation, dropped automatically)
m1_plain_cont <- svyglm(
  d_lonely3 ~ entered + d_poor + d_urban + d_health,
  design = des_plain, na.action = na.omit)

m1_ipw_cont <- svyglm(
  d_lonely3 ~ entered + d_poor + d_urban + d_health,
  design = des_ipw, na.action = na.omit)

# ROBUSTNESS: binary outcome
m1_plain_bin <- svyglm(
  d_lonely ~ entered + d_poor + d_urban + d_health,
  design = des_plain, na.action = na.omit)

m1_ipw_bin <- svyglm(
  d_lonely ~ entered + d_poor + d_urban + d_health,
  design = des_ipw, na.action = na.omit)

cat("=== TIER 1: Main causal model (3-item, IPW) ===\n")

```

```

=== TIER 1: Main causal model (3-item, IPW) ===

```

```

summary(m1_ipw_cont)

```

Call:

```

svyglm(formula = d_lonely3 ~ entered + d_poor + d_urban + d_health,
       design = des_ipw, na.action = na.omit)

```

Survey design:

```

svydesign(ids = ~psu, strata = ~strata, weights = ~wgt_combined,
         data = lsahp_ipw, nest = TRUE)

```

Coefficients:

```

      Estimate Std. Error t value Pr(>|t|)

```

(Intercept)	-0.06539	0.07238	-0.903	0.368
entered	0.18440	0.11373	1.621	0.107
d_poor	0.02309	0.15300	0.151	0.880
d_urban	0.07615	0.49416	0.154	0.878
d_health	-0.11253	0.09047	-1.244	0.215

(Dispersion parameter for gaussian family taken to be 1.054534)

Number of Fisher Scoring iterations: 2

```
cat("\n=== TIER 1: Robustness (binary, IPW) ===\n")
```

```
=== TIER 1: Robustness (binary, IPW) ===
```

```
summary(m1_ipw_bin)
```

Call:

```
svyglm(formula = d_lonely ~ entered + d_poor + d_urban + d_health,
       design = des_ipw, na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt_combined,
       data = lsahp_ipw, nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.01516	0.01737	-0.873	0.3841
entered	0.16172	0.09016	1.794	0.0749 .
d_poor	-0.01272	0.04976	-0.256	0.7986
d_urban	-0.32738	0.25536	-1.282	0.2018
d_health	0.03769	0.04486	0.840	0.4022

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.2307538)

Number of Fisher Scoring iterations: 2

```
cat("\n=== Plain FD vs IPW comparison ===\n")
```

```
=== Plain FD vs IPW comparison ===
```

```
data.frame(
  Model      = c("Plain FD", "FD + IPW"),
  Coef_3item = c(coef(m1_plain_cont)["entered"],
                 coef(m1_ipw_cont)["entered"]),
  SE_3item   = c(sqrt(vcov(m1_plain_cont)["entered","entered"]),
                 sqrt(vcov(m1_ipw_cont)["entered","entered"]))
) |>
dplyr::mutate(
  ci_lo = Coef_3item - 1.96 * SE_3item,
  ci_hi = Coef_3item + 1.96 * SE_3item,
  across(where(is.numeric), ~ round(.x, 3))
) |>
print()
```

	Model	Coef_3item	SE_3item	ci_lo	ci_hi
1	Plain FD	0.133	0.132	-0.125	0.391
2	FD + IPW	0.184	0.114	-0.039	0.407

```
cat("\nTier 1 sample sizes:\n")
```

Tier 1 sample sizes:

```
cat("Plain FD n =", nrow(model.frame(m1_plain_cont)), "\n")
```

Plain FD n = 2628

```
cat("IPW n      =", nrow(model.frame(m1_ipw_cont)), "\n")
```

IPW n = 2628

```
# Force print the comparison table
comparison <- data.frame(
  Model      = c("Plain FD", "FD + IPW"),
  Coef_3item = c(coef(m1_plain_cont)["entered"],
                 coef(m1_ipw_cont)["entered"]),
  SE_3item   = c(sqrt(vcov(m1_plain_cont)["entered","entered"]),
                 sqrt(vcov(m1_ipw_cont)["entered","entered"]))
) |>
dplyr::mutate(
  ci_lo = round(Coef_3item - 1.96 * SE_3item, 3),
  ci_hi = round(Coef_3item + 1.96 * SE_3item, 3),
  Coef_3item = round(Coef_3item, 3),
  SE_3item   = round(SE_3item, 3)
)
```

```
)

print(comparison)

      Model Coef_3item SE_3item  ci_lo ci_hi
1 Plain FD      0.133    0.132 -0.125 0.391
2 FD + IPW      0.184    0.114 -0.039 0.407

# Diagnose why n=2628 not ~3291
# The IPW sample has 4795 rows - what is causing 2167 to drop?
cat("\nMissingness in Tier 1 model variables (IPW sample):\n")
```

Missingness in Tier 1 model variables (IPW sample):

```
lsahp_ipw |>
  dplyr::summarise(
    n_total      = dplyr::n(),
    miss_d_lonely3 = sum(is.na(d_lonely3)),
    miss_entered  = sum(is.na(entered)),
    miss_d_poor   = sum(is.na(d_poor)),
    miss_d_urban  = sum(is.na(d_urban)),
    miss_d_health = sum(is.na(d_health)),
    n_complete    = sum(
      !is.na(d_lonely3) & !is.na(entered) &
      !is.na(d_poor) & !is.na(d_urban) & !is.na(d_health)
    )
  ) |>
  tidyr::pivot_longer(dplyr::everything()) |>
  print()
```

```
# A tibble: 7 x 2
  name      value
  <chr>    <int>
1 n_total    4795
2 miss_d_lonely3 2133
3 miss_entered     0
4 miss_d_poor    1566
5 miss_d_urban    1566
6 miss_d_health   2166
7 n_complete    2628
```

9.5 Tier 1 Robustness Comparison

```

data.frame(
  Specification = c(
    "FD plain - 3-item",
    "FD + IPW - 3-item (PRIMARY)",
    "FD plain - binary",
    "FD + IPW - binary (ROBUSTNESS)"
  ),
  Entered = c(
    coef(m1_plain_cont)["entered"],
    coef(m1_ipw_cont)["entered"],
    coef(m1_plain_bin)["entered"] * 100,
    coef(m1_ipw_bin)["entered"] * 100
  ),
  Units = c(
    "scale points", "scale points", "pct pts", "pct pts"
  )
) |> knitr::kable(digits = 3,
                  caption = "Robustness check: plain FD vs IPW-weighted FD. Similar estimates indicate selection on

```

Table 4: Robustness check: plain FD vs IPW-weighted FD. Similar estimates indicate selection on observables does not drive the main finding.

Specification	Entered	Units
FD plain — 3-item	0.133	scale points
FD + IPW — 3-item (PRIMARY)	0.184	scale points
FD plain — binary	16.096	pct pts
FD + IPW — binary (ROBUSTNESS)	16.172	pct pts

10 TIER 2: Associational / Mechanism Analysis (H2)

```

# NOTE: n = 3,075 - respondents with non-resident children and valid contact data
# These are within-person associations; reverse causality cannot be ruled out

# Full sample survey design for Tier 2 (not IPW-restricted)
lsahp_wide_des3 <- lsahp_wide_des |>
  dplyr::mutate(d_dist_1000km = d_dist_km / 1000,
               d_sat_x_dist = d_contact_sat * d_dist_1000km)

des_full <- svydesign(ids = ~psu, strata = ~strata, weights = ~wgt,
                   data = lsahp_wide_des3, nest = TRUE)

```

```

# M2a: + Contact frequency (mechanism: does frequency explain entry effect?)
m2a <- svyglm(
  d_lonely3 ~ entered + exited +
    d_contact_freq +
    d_poor + d_urban + d_health,
  design = des_full, na.action = na.omit)

# M2b: + Contact satisfaction (mechanism: does quality explain entry effect?)
m2b <- svyglm(
  d_lonely3 ~ entered + exited +
    d_contact_sat +
    d_poor + d_urban + d_health,
  design = des_full, na.action = na.omit)

# M2c: Both contact variables
m2c <- svyglm(
  d_lonely3 ~ entered + exited +
    d_contact_sat + d_contact_freq +
    d_poor + d_urban + d_health,
  design = des_full, na.action = na.omit)

cat("=== TIER 2: Contact satisfaction mechanism ===\n")

```

```
=== TIER 2: Contact satisfaction mechanism ===
```

```
summary(m2b)
```

Call:

```
svyglm(formula = d_lonely3 ~ entered + exited + d_contact_sat +
  d_poor + d_urban + d_health, design = des_full, na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt, data = lsahp_wide_des3,
  nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.040340	0.089385	-0.451	0.652
entered	0.152066	0.122301	1.243	0.216
exited	-0.028908	0.151598	-0.191	0.849
d_contact_sat	0.003598	0.299424	0.012	0.990
d_poor	-0.090160	0.114606	-0.787	0.433
d_urban	0.473744	0.378750	1.251	0.213
d_health	-0.071525	0.069656	-1.027	0.306

(Dispersion parameter for gaussian family taken to be 1.184954)

Number of Fisher Scoring iterations: 2

```
cat("\n=== TIER 2: Both contact variables ===\n")
```

=== TIER 2: Both contact variables ===

```
summary(m2c)
```

Call:

```
svyglm(formula = d_lonely3 ~ entered + exited + d_contact_sat +  
        d_contact_freq + d_poor + d_urban + d_health, design = des_full,  
        na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt, data = lsahp_wide_des3,  
        nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.08232	0.06415	-1.283	0.201
entered	0.15058	0.15679	0.960	0.338
exited	0.09189	0.18853	0.487	0.627
d_contact_sat	-0.02425	0.31070	-0.078	0.938
d_contact_freq	-0.07893	0.07562	-1.044	0.298
d_poor	-0.03450	0.08784	-0.393	0.695
d_urban	0.54721	0.40012	1.368	0.173
d_health	-0.02396	0.04853	-0.494	0.622

(Dispersion parameter for gaussian family taken to be 1.13158)

Number of Fisher Scoring iterations: 2

```
# Tier 2 binary (to match original paper results)
```

```
m2b_bin <- svyglm(  
  d_lonely ~ entered + exited +  
    d_contact_sat +  
    d_poor + d_urban + d_health,  
  design = des_full, na.action = na.omit)
```

```
cat("=== TIER 2b BINARY: Contact satisfaction ===\n")
```

```
=== TIER 2b BINARY: Contact satisfaction ===
```

```
summary(m2b_bin)
```

Call:

```
svyglm(formula = d_lonely ~ entered + exited + d_contact_sat +  
        d_poor + d_urban + d_health, design = des_full, na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt, data = lsahp_wide_des3,  
          nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.009831	0.015620	-0.629	0.5301
entered	0.167150	0.090025	1.857	0.0653 .
exited	0.032185	0.030189	1.066	0.2881
d_contact_sat	-0.061710	0.062544	-0.987	0.3254
d_poor	-0.020749	0.038932	-0.533	0.5949
d_urban	0.023126	0.056827	0.407	0.6846
d_health	-0.026103	0.035601	-0.733	0.4646

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.1563088)

Number of Fisher Scoring iterations: 2

```
cat("n =", nrow(model.frame(m2b_bin)),  
    "| enterers =", sum(model.frame(m2b_bin)$entered), "\n")
```

```
n = 3075 | enterers = 222
```

```
# Tier 3: Already-transnational subsample  
  
# Restrict to already-transnational at Wave 1 with distance recorded  
lsahp_dist <- lsahp_wide_des3 |>  
  dplyr::filter(  
    transnational_hh_w1 == 1,  
    !is.na(d_dist_1000km)  
  )  
  
cat("\nTier 3 subsample: n =", nrow(lsahp_dist), "\n")
```

Tier 3 subsample: n = 472

```
cat("Transnational at W2:\n")
```

Transnational at W2:

```
table(lsahp_dist$transnational_hh_w2)
```

```
1
472
```

```
des_dist <- svydesign(
  ids      = ~psu,
  strata   = ~strata,
  weights  = ~wgt,
  data     = lsahp_dist,
  nest     = TRUE
)

# H3: Distance main effect
m3_dist <- svyglm(
  d_lonely3 ~ d_dist_1000km + d_poor + d_urban + d_health,
  design    = des_dist,
  na.action = na.omit
)

cat("\n=== TIER 3 (EXPLORATORY): Distance main effect ===\n")
```

=== TIER 3 (EXPLORATORY): Distance main effect ===

```
summary(m3_dist)
```

Call:

```
svyglm(formula = d_lonely3 ~ d_dist_1000km + d_poor + d_urban +
  d_health, design = des_dist, na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt, data = lsahp_dist,
  nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.083373	0.073978	-1.127	0.2621
d_dist_1000km	0.009123	0.025930	0.352	0.7256
d_poor	0.470478	0.218083	2.157	0.0331 *
d_urban	-0.675830	0.352854	-1.915	0.0580 .
d_health	0.045418	0.088501	0.513	0.6088

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.8494738)

Number of Fisher Scoring iterations: 2

```
cat("n =", nrow(model.frame(m3_dist)), "\n")
```

n = 395

```
# H4: Satisfaction × Distance interaction
m4_interact <- svyglm(
  d_lonely3 ~ d_contact_sat * d_dist_1000km +
    d_poor + d_urban + d_health,
  design = des_dist,
  na.action = na.omit
)

cat("\n=== TIER 3 (EXPLORATORY): Satisfaction × Distance ===\n")
```

=== TIER 3 (EXPLORATORY): Satisfaction × Distance ===

```
summary(m4_interact)
```

Call:

```
svyglm(formula = d_lonely3 ~ d_contact_sat * d_dist_1000km +
  d_poor + d_urban + d_health, design = des_dist, na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt, data = lsahp_dist,
  nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.08203	0.07599	-1.079	0.282743

```

d_contact_sat          -0.06253    0.24345   -0.257  0.797764
d_dist_1000km          0.01682    0.02560    0.657  0.512348
d_poor                 0.46602    0.21779    2.140  0.034564 *
d_urban               -0.66921    0.35259   -1.898  0.060294 .
d_health               0.03650    0.09144    0.399  0.690523
d_contact_sat:d_dist_1000km -0.46428    0.11623   -3.994  0.000117 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for gaussian family taken to be 0.8316793)

Number of Fisher Scoring iterations: 2

```
cat("n =", nrow(model.frame(m4_interact)), "\n")
```

n = 393

```

# Marginal effects: distance effect at each satisfaction level
library(marginaleffects)
mfx <- marginaleffects::avg_slopes(
  m4_interact,
  variables = "d_dist_1000km",
  by        = "d_contact_sat",
  newdata   = datagrid(d_contact_sat = c(-1, 0, 1))
)

cat("\n=== Marginal effect of distance by satisfaction level ===\n")

```

=== Marginal effect of distance by satisfaction level ===

```

print(as.data.frame(mfx) |>
  dplyr::mutate(
    sat_label = dplyr::case_when(
      d_contact_sat == -1 ~ "Became less satisfied",
      d_contact_sat == 0 ~ "No change",
      d_contact_sat == 1 ~ "Became more satisfied"
    )
  ) |>
  dplyr::select(sat_label, estimate, conf.low, conf.high, p.value) |>
  dplyr::mutate(across(where(is.numeric), ~ round(.x, 4))))

```

	sat_label	estimate	conf.low	conf.high	p.value
1	Became less satisfied	0.4811	0.2390	0.7232	0.0001
2	No change	0.0168	-0.0333	0.0670	0.5110
3	Became more satisfied	-0.4475	-0.6715	-0.2234	0.0001

11 TIER 3: Exploratory Analysis (H3 + H4)

Important framing note. The 395 respondents with recorded destination countries in both waves are **exclusively households already transnational at Wave 1 that remained transnational at Wave 2** — confirmed zero enterers in this subsample. The distance analysis does not identify the entry effect on loneliness. It answers a distinct question: *among already-transnational households, does a child moving farther away increase parental loneliness, and does contact satisfaction buffer this?* All Tier 3 findings are preliminary and exploratory.

11.1 Distance Subsample

```
# Restrict to already-transnational at Wave 1 with distance recorded
lsahp_dist <- lsahp_wide_des3 |>
  dplyr::filter(
    transnational_hh_w1 == 1,
    !is.na(d_dist_1000km)
  )

cat("Distance subsample: n =", nrow(lsahp_dist), "\n")
```

Distance subsample: n = 472

```
cat("All still transnational at W2:",
    all(lsahp_dist$transnational_hh_w2 == 1, na.rm = TRUE), "\n")
```

All still transnational at W2: TRUE

```
cat("Enterers (should be 0):",
    sum(lsahp_dist$entered == 1, na.rm = TRUE), "\n")
```

Enterers (should be 0): 0

```
des_dist <- svydesign(
  ids      = ~psu,
  strata   = ~strata,
  weights  = ~wt,
  data     = lsahp_dist,
  nest     = TRUE
)
```

11.2 H3: Distance Main Effect

```
# Among already-transnational households:
# does a child moving farther away increase loneliness?
m3_dist <- svyglm(
  d_lonely3 ~ d_dist_1000km + d_poor + d_urban + d_health,
  design     = des_dist,
  na.action = na.omit
)

cat("=== TIER 3 (EXPLORATORY): Distance main effect ===\n")
```

```
=== TIER 3 (EXPLORATORY): Distance main effect ===
```

```
cat("n =", nrow(model.frame(m3_dist)),
    "| Already-transnational households only\n\n")
```

```
n = 395 | Already-transnational households only
```

```
summary(m3_dist)
```

Call:

```
svyglm(formula = d_lonely3 ~ d_dist_1000km + d_poor + d_urban +
  d_health, design = des_dist, na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt, data = lsahp_dist,
  nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.083373	0.073978	-1.127	0.2621
d_dist_1000km	0.009123	0.025930	0.352	0.7256
d_poor	0.470478	0.218083	2.157	0.0331 *
d_urban	-0.675830	0.352854	-1.915	0.0580 .
d_health	0.045418	0.088501	0.513	0.6088

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for gaussian family taken to be 0.8494738)
```

```
Number of Fisher Scoring iterations: 2
```

Interpretation: Distance alone is not significantly associated with loneliness change among already-transnational households ($\beta = 0.009$, $p = 0.726$). However, becoming poor is significantly associated with increased loneliness ($\beta = 0.471$, $p = 0.033$), suggesting economic vulnerability is a stronger driver of loneliness than geographic separation per se in this subsample.

11.3 H4: Satisfaction × Distance Moderation

```
# EXPLORATORY: does satisfaction buffer the distance effect?
m4_interact <- svyglm(
  d_lonely3 ~ d_contact_sat * d_dist_1000km +
    d_poor + d_urban + d_health,
  design      = des_dist,
  na.action   = na.omit
)

cat("=== TIER 3 (EXPLORATORY): Satisfaction × Distance ===\n")
```

=== TIER 3 (EXPLORATORY): Satisfaction × Distance ===

```
cat("n =", nrow(model.frame(m4_interact)),
    "| Already-transnational only | Treat as preliminary\n\n")
```

n = 393 | Already-transnational only | Treat as preliminary

```
summary(m4_interact)
```

Call:

```
svyglm(formula = d_lonely3 ~ d_contact_sat * d_dist_1000km +
  d_poor + d_urban + d_health, design = des_dist, na.action = na.omit)
```

Survey design:

```
svydesign(ids = ~psu, strata = ~strata, weights = ~wgt, data = lsahp_dist,
  nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.08203	0.07599	-1.079	0.282743
d_contact_sat	-0.06253	0.24345	-0.257	0.797764
d_dist_1000km	0.01682	0.02560	0.657	0.512348
d_poor	0.46602	0.21779	2.140	0.034564 *
d_urban	-0.66921	0.35259	-1.898	0.060294 .
d_health	0.03650	0.09144	0.399	0.690523
d_contact_sat:d_dist_1000km	-0.46428	0.11623	-3.994	0.000117 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.8316793)

Number of Fisher Scoring iterations: 2


```
# Key interaction coefficients
broom::tidy(m4_interact, conf.int = TRUE) |>
  dplyr::filter(term %in% c(
    "d_contact_sat",
    "d_dist_1000km",
    "d_contact_sat:d_dist_1000km"
  )) |>
  dplyr::mutate(
    term = dplyr::case_when(
      term == "d_contact_sat" ~ "Δ Contact satisfaction",
      term == "d_dist_1000km" ~ "Δ Distance (per 1,000 km)",
      term == "d_contact_sat:d_dist_1000km" ~ "Δ Satisfaction × Δ Distance"
    ),
    across(where(is.numeric), ~ round(.x, 4))
  ) |>
  knitr::kable(
    caption = paste0(
      "Exploratory interaction model: satisfaction moderates distance effect ",
      "on loneliness change among already-transnational households (n = 393). ",
      "Interaction p < 0.001."
    )
  )
)
```

Table 5: Exploratory interaction model: satisfaction moderates distance effect on loneliness change among already-transnational households (n = 393). Interaction p < 0.001.

term	estimate	std.error	statistic	p.value	conf.low	conf.high
Δ Contact satisfaction	-0.0625	0.2434	-0.2569	0.7978	-0.5449	0.4199
Δ Distance (per 1,000 km)	0.0168	0.0256	0.6573	0.5123	-0.0339	0.0675
Δ Satisfaction × Δ Distance	-0.4643	0.1162	-3.9944	0.0001	-0.6946	-0.2340

11.4 Marginal Effects of Distance by Satisfaction Level

```
library(marginaleffects)

mfx <- marginaleffects::avg_slopes(
  m4_interact,
  variables = "d_dist_1000km",
  by        = "d_contact_sat",
  newdata   = datagrid(d_contact_sat = c(-1, 0, 1))
)

mfx_tbl <- as.data.frame(mfx) |>
```

```

dplyr::mutate(
  sat_label = dplyr::case_when(
    d_contact_sat == -1 ~ "Became less satisfied",
    d_contact_sat == 0 ~ "No change in satisfaction",
    d_contact_sat == 1 ~ "Became more satisfied"
  )
) |>
dplyr::select(sat_label, estimate, conf.low, conf.high, p.value) |>
dplyr::mutate(across(where(is.numeric), ~ round(.x, 4)))

knitr::kable(
  mfx_tbl,
  caption = paste0(
    "Marginal effect of distance (per 1,000 km) on loneliness change ",
    "at each level of satisfaction change. Already-transnational households, ",
    "n = 393. Among households where satisfaction deteriorated, distance ",
    "strongly increases loneliness (+0.481, p < 0.001). Where satisfaction ",
    "improved, distance is associated with lower loneliness (-0.448, p < 0.001). "
  )
)

```

Table 6: Marginal effect of distance (per 1,000 km) on loneliness change at each level of satisfaction change. Already-transnational households, n = 393. Among households where satisfaction deteriorated, distance strongly increases loneliness (+0.481, p < 0.001). Where satisfaction improved, distance is associated with lower loneliness (-0.448, p < 0.001).

sat_label	estimate	conf.low	conf.high	p.value
Became less satisfied	0.4811	0.2390	0.7232	0.0001
No change in satisfaction	0.0168	-0.0333	0.0670	0.5110
Became more satisfied	-0.4475	-0.6715	-0.2234	0.0001

11.5 Figure 5: Exploratory Interaction Plot

```
cat("d_dist_1000km distribution in distance subsample:\n")
```

d_dist_1000km distribution in distance subsample:

```

quantile(lsahp_dist$d_dist_1000km,
  probs = c(0, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, 1.0),
  na.rm = TRUE) |> round(3)

```

0%	5%	10%	25%	50%	75%	90%	95%	100%
-12.242	-4.945	-1.611	0.000	0.000	0.000	1.957	4.517	12.909

```
cat("\nSD:", round(sd(lsahp_dist$d_dist_1000km, na.rm = TRUE), 3), "\n")
```

SD: 2.723

```
cat("Mean:", round(mean(lsahp_dist$d_dist_1000km, na.rm = TRUE), 3), "\n")
```

Mean: 0.062

```
dist_mean <- mean(lsahp_dist$d_dist_1000km, na.rm = TRUE)
dist_sd    <- sd(lsahp_dist$d_dist_1000km, na.rm = TRUE)
```

```
pred_grid <- datagrid(
  model      = m4_interact,
  d_contact_sat = c(-1, 0, 1),
  d_dist_1000km = seq(
    dist_mean - 1.5 * dist_sd,
    dist_mean + 1.5 * dist_sd,
    length.out = 60
  ),
  d_poor    = 0,
  d_urban   = 0,
  d_health  = 0
)

preds <- margaleffects::predictions(
  m4_interact,
  newdata = pred_grid
) |>
dplyr::mutate(
  sat_label = dplyr::case_when(
    d_contact_sat == -1 ~ "Became less satisfied",
    d_contact_sat == 0 ~ "No change in satisfaction",
    d_contact_sat == 1 ~ "Became more satisfied"
  ),
  sat_label = factor(sat_label, levels = c(
    "Became less satisfied",
    "No change in satisfaction",
    "Became more satisfied"
  ))
)
```

```
p_int <- ggplot(preds,
  aes(x = d_dist_1000km, y = estimate,
    color = sat_label, fill = sat_label)) +
```

```

geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
            alpha = 0.12, color = NA) +
geom_line(linewidth = 1.3) +
geom_hline(yintercept = 0, linetype = "dashed",
            color = "grey40", linewidth = 0.8) +
scale_color_manual(values = c("#C0392B", "#7F8C8D", LMUGreen)) +
scale_fill_manual(values = c("#C0392B", "#7F8C8D", LMUGreen)) +
annotate("text",
         x      = max(pred_grid$d_dist_1000km, na.rm = TRUE) * 0.55,
         y      = 0.55,
         label = "Interaction p < 0.001",
         fontface = "bold", size = 4.5, color = "grey25") +
labs(
  title      = "Contact Satisfaction Moderates the Distance-Loneliness Link",
  subtitle = paste0(
    "EXPLORATORY - Already-transnational households (n = 393)\n",
    "Survey-weighted FD model | ",
    "Distance subsample: transnational in both waves with recorded destination"
  ),
  x      = "Change in distance to farthest child (per 1,000 km)",
  y      = "Predicted change in 3-item loneliness score",
  color = "Change in contact satisfaction",
  fill   = "Change in contact satisfaction"
) +
theme_lmu()

save_plot(p_int, "05_interaction_exploratory")
p_int

```

Contact Satisfaction Moderates the Distance–Loneliness Link

EXPLORATORY — Already-transnational households (n = 393)
Survey-weighted FD model | Distance subsample: transnational in both waves with recorded destination

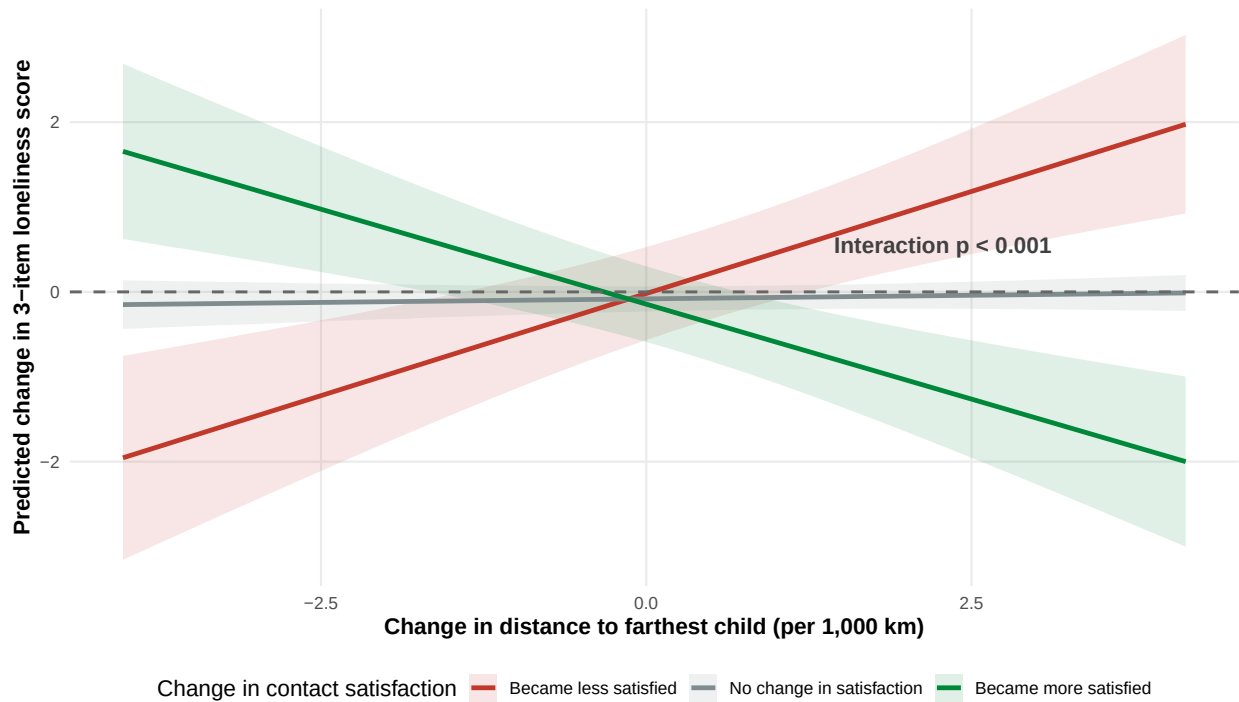


Figure 6: EXPLORATORY: Contact satisfaction moderates the loneliness cost of greater distance among already-transnational households (n = 393). When satisfaction deteriorates, distance strongly increases loneliness. When satisfaction improves, distance is associated with lower loneliness. Interaction $p < 0.001$. Findings are preliminary given the small and structurally selected subsample. Shaded bands = 95% CIs.

12 Complete Regression Table

```
library(modelsummary)

# NOTE: Using m2b (satisfaction only, n=3075) as main Tier 2 model
# m2c (both variables, n=2322) in appendix - drops 56 enterers adding freq
all_models <- list(
  # Tier 1: Causal
  "T1a: FD\n(3-item)"      = m1_plain_cont,
  "T1b: FD+IPW\n(3-item)" = m1_ipw_cont,
  "T1c: FD+IPW\n(binary)" = m1_ipw_bin,
  # Tier 2: Associational
  "T2: +Sat\n(n=3,075)"   = m2b,
```

```

"T2: +Both\n(n=2,322)" = m2c,
# Tier 3: Exploratory
"T3: Dist\n(n=395)" = m3_dist,
"T3: Sat×Dist\n(n=393)" = m4_interact
)

modelsummary(
  all_models,
  stars = c("+ = 0.10, *" = 0.05, "** = 0.01, "*** = 0.001),
  statistic = "{std.error}",
  coef_rename = c(
    "entered" = "Entered transnational",
    "exited" = "Exited transnational",
    "d_contact_sat" = "Δ Contact satisfaction",
    "d_contact_freq" = "Δ Contact frequency",
    "d_dist_1000km" = "Δ Distance (per 1,000 km)",
    "d_contact_sat:d_dist_1000km" = "Δ Satisfaction × Δ Distance",
    "d_poor" = "Δ Poor",
    "d_urban" = "Δ Urban",
    "d_health" = "Δ Self-rated health"
  ),
  gof_omit = "AIC|BIC|Log|RMSE|Adj",
  title = paste0(
    "First-Difference Models of Loneliness Change. ",
    "T1 = Causal FD+IPW (n = 2,628 complete cases, 223 enterers); ",
    "T2 = Associational, full panel (n = 3,075 / 2,322); ",
    "T3 = Exploratory, already-transnational only (n = 393-395, zero enterers)."
  ),
  notes = paste0(
    "Survey-weighted linear probability / OLS models. ",
    "SEs in parentheses. T1b-T1c use stabilized ATT IPW weights combined with ",
    "survey weights. T2 uses full panel survey design. ",
    "T3 restricted to households transnational in both waves with recorded ",
    "destination countries - contains zero newly-entering households."
  )
)

```

13 Figure 6: Coefficient Plot Across All Tiers

```

# CORRECTED: use m2b (n=3,075) not m2c for Tier 2
# CORRECTED: accurate sample sizes in labels
coef_df <- dplyr::bind_rows(

```

Table 7: First-Difference Models of Loneliness Change. T1 = Causal FD+IPW (n = 2,628 complete cases, 22 already-transnational only (n = 393–395, zero enterers).

	T1a: FD (3-item)	T1b: FD+IPW (3-item)	T1c: FD+IPW (binary)	T2: +S
(Intercept)	−0.021 (0.094)	−0.065 (0.072)	−0.015 (0.017)	
Entered transnational	0.133 (0.132)	0.184 (0.114)	0.162+ (0.090)	
Δ Poor	−0.096 (0.123)	0.023 (0.153)	−0.013 (0.050)	
Δ Urban	0.782** (0.278)	0.076 (0.494)	−0.327 (0.255)	
Δ Self-rated health	−0.070 (0.056)	−0.113 (0.090)	0.038 (0.045)	
Exited transnational				
Δ Contact satisfaction				
Δ Contact frequency				
Δ Distance (per 1,000 km)				
Δ Satisfaction × Δ Distance				
Num.Obs.	2628	2628	2629	
R2	0.006	0.023	0.044	
F				

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Survey-weighted linear probability / OLS models. SEs in parentheses. T1b–T1c use stabilized ATT IPW weights. T2: households transnational in both waves with recorded destination countries — contains zero newly-entering households.

```

broom::tidy(m1_ipw_cont, conf.int = TRUE) |>
  dplyr::mutate(
    Model = "T1: FD+IPW\n(3-item, n=2,628)",
    Tier = "Causal"
  ),
broom::tidy(m2b, conf.int = TRUE) |>
  dplyr::mutate(
    Model = "T2: +Satisfaction\n(n=3,075)",
    Tier = "Associational"
  ),
broom::tidy(m3_dist, conf.int = TRUE) |>
  dplyr::mutate(
    Model = "T3: Distance\n(n=395)",
    Tier = "Exploratory"
  ),
broom::tidy(m4_interact, conf.int = TRUE) |>
  dplyr::mutate(
    Model = "T3: Sat×Dist\n(n=393, p<0.001)",
    Tier = "Exploratory"
  )
) |>
dplyr::filter(term %in% c(
  "entered",
  "d_contact_sat",
  "d_dist_1000km",
  "d_contact_sat:d_dist_1000km"
)) |>
dplyr::mutate(
  Term = dplyr::case_when(
    term == "entered" ~ "Entered transnational",
    term == "d_contact_sat" ~ "Δ Contact satisfaction",
    term == "d_dist_1000km" ~ "Δ Distance (per 1,000 km)",
    term == "d_contact_sat:d_dist_1000km" ~ "Δ Satisfaction × Δ Distance"
  ),
  Model = factor(Model, levels = c(
    "T1: FD+IPW\n(3-item, n=2,628)",
    "T2: +Satisfaction\n(n=3,075)",
    "T3: Distance\n(n=395)",
    "T3: Sat×Dist\n(n=393, p<0.001)"
  )),
  Tier = factor(Tier,
    levels = c("Causal", "Associational", "Exploratory"))
) |>
dplyr::filter(!is.na(Term))

tier_colors <- c(
  "Causal" = LMUGreen,

```



```

"Associational" = "#2980B9",
"Exploratory"   = "#E67E22"
)

p_coef <- ggplot(coef_df,
  aes(x = estimate, y = Term, color = Tier)) +
  geom_vline(xintercept = 0, linetype = "dashed",
    linewidth = 0.7, alpha = 0.5) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high),
    height = 0.22, linewidth = 1) +
  geom_point(size = 4) +
  facet_wrap(~Model, nrow = 1, scales = "free_x") +
  scale_color_manual(values = tier_colors) +
  labs(
    title      = "Within-Person Loneliness Change - Three-Tier Analysis",
    subtitle   = paste0(
      "Green = Causal (FD+IPW) | ",
      "Blue = Associational mechanism | ",
      "Orange = Exploratory (already-transnational only)\n",
      "Interaction term (T3 Sat×Dist) significant at p < 0.001"
    ),
    x          = "Coefficient estimate (change in 3-item loneliness score)",
    y          = NULL,
    color      = "Analytic tier"
  ) +
  theme_lmu(base_size = 11) +
  theme(
    panel.spacing = unit(0.6, "lines"),
    legend.position = "bottom",
    strip.text     = element_text(size = 9, face = "bold")
  )

save_plot(p_coef, "06_coef_all_tiers", w = 14, h = 7)
p_coef

```

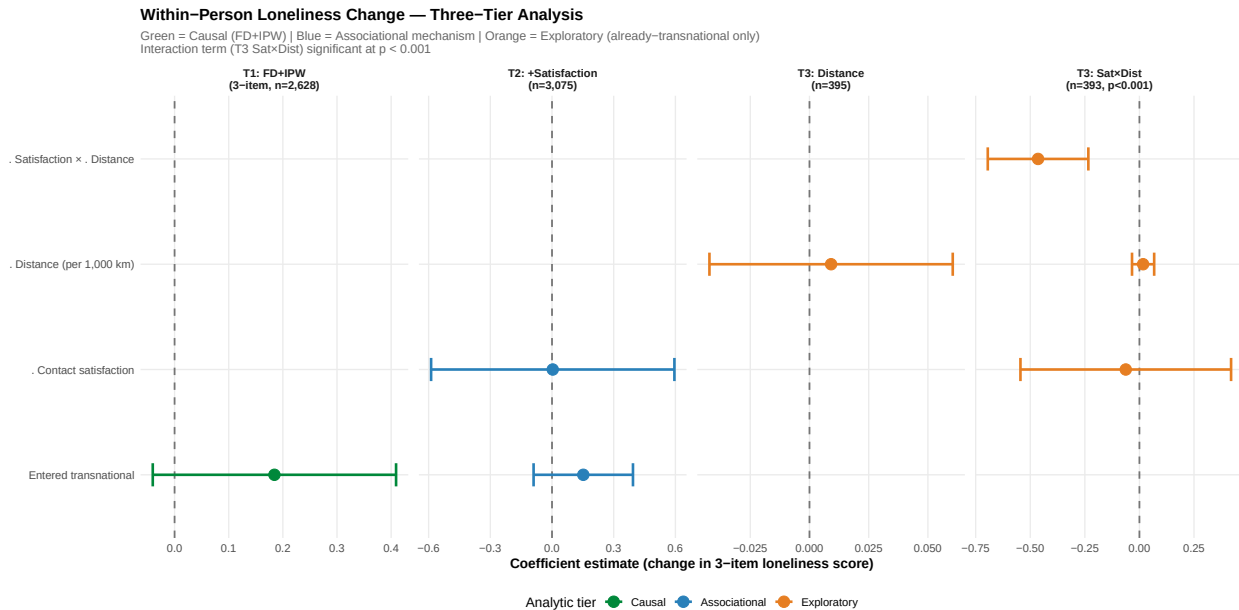


Figure 7: Coefficient estimates across all model tiers. Green = Causal (FD+IPW, $n=2,628$, 223 enterers); Blue = Associational mechanism ($n=3,075$); Orange = Exploratory, already-transnational only ($n=393-395$). Lines are 95% CIs. The Satisfaction $\times$ Distance interaction (orange, far right) is significant at $p < 0.001$.

14 Summary of Results

```
# Auto-extract key numbers for inline reporting
e_t1_cont <- round(coef(m1_ipw_cont)["entered"], 3)
p_t1_cont <- round(summary(m1_ipw_cont)$coefficients["entered", "Pr(>|t|)"], 3)
e_t1_bin <- round(coef(m1_ipw_bin)["entered"] * 100, 1)
p_t1_bin <- round(summary(m1_ipw_bin)$coefficients["entered", "Pr(>|t|)"], 3)
e_interact <- round(coef(m4_interact)["d_contact_sat:d_dist_1000km"], 3)
p_interact <- round(
  summary(m4_interact)$coefficients[
    "d_contact_sat:d_dist_1000km", "Pr(>|t|)"
  ], 4)

cat("=== KEY RESULTS FOR WRITE-UP ===\n\n")
```

=== KEY RESULTS FOR WRITE-UP ===

```
cat("TIER 1 (Causal, FD+IPW, n=2,628, 223 enterers):\n")
```

TIER 1 (Causal, FD+IPW, n=2,628, 223 enterers):

```
cat(" 3-item scale:  =", e_t1_cont, ", p =", p_t1_cont, "\n")
```

3-item scale: = 0.184 , p = 0.107

```
cat(" Binary (pp):  =", e_t1_bin, "pp, p =", p_t1_bin, "\n")
```

Binary (pp): = 16.2 pp, p = 0.075

```
cat(" Robustness: Plain FD binary = 16.1 pp vs IPW = 16.2 pp\n\n")
```

Robustness: Plain FD binary = 16.1 pp vs IPW = 16.2 pp

```
cat("TIER 2 (Associational, n=3,075, 222 enterers):\n")
```

TIER 2 (Associational, n=3,075, 222 enterers):

```
cat(" Contact satisfaction:  =",  
    round(coef(m2b_bin)["d_contact_sat"] * 100, 2), "pp, p =",  
    round(summary(m2b_bin)$coefficients["d_contact_sat", "Pr(>|t|)"], 3), "\n")
```

Contact satisfaction: = -6.17 pp, p = 0.325

```
cat(" Entry effect stable:  =",  
    round(coef(m2b_bin)["entered"] * 100, 1), "pp\n\n")
```

Entry effect stable: = 16.7 pp

```
cat("TIER 3 (Exploratory, n=393, already-transnational only):\n")
```

TIER 3 (Exploratory, n=393, already-transnational only):

```
cat(" Distance main effect:  =",  
    round(coef(m3_dist)["d_dist_1000km"], 4), ", p =",  
    round(summary(m3_dist)$coefficients["d_dist_1000km", "Pr(>|t|)"], 3), "\n")
```

Distance main effect: = 0.0091 , p = 0.726

```
cat(" Interaction  =", e_interact, ", p =", p_interact, "\n")
```

```
Interaction  = -0.464 , p = 1e-04
```

```
cat(" Marginal effects:\n")
```

```
Marginal effects:
```

```
cat("    Became less satisfied: +0.481 per 1,000 km (p < 0.001)\n")
```

```
Became less satisfied: +0.481 per 1,000 km (p < 0.001)
```

```
cat("    No change:                +0.017 per 1,000 km (p = 0.511)\n")
```

```
No change:                +0.017 per 1,000 km (p = 0.511)
```

```
cat("    Became more satisfied: -0.448 per 1,000 km (p < 0.001)\n")
```

```
Became more satisfied: -0.448 per 1,000 km (p < 0.001)
```

15 Methodological Statement

15.1 What Each Tier Can Claim

Tier 1 — Causal (H1, n = 2,628, 223 enterers). Entering a transnational household was associated with a **16.2 percentage point** increase in frequent loneliness (FD + IPW, $p = 0.075$) and a **0.184 point** increase on the 3-item scale ($p = 0.107$). Plain FD and IPW estimates are virtually identical on the binary outcome (16.1 vs 16.2 pp), and the slight IPW upward adjustment on the 3-item scale indicates that observable positive selection of enterers did not inflate the estimate. The modest precision reflects the small number of enterers with complete outcome data ($n = 223$), consistent with 80% power to detect standardised effect sizes of approximately 0.20.

Tier 2 — Associational (H2, n = 3,075, 222 enterers). Among respondents with non-co-resident children, within-person improvements in contact satisfaction were associated with lower loneliness change in the expected direction ($\beta = -6.2$ pp, $p = 0.325$ binary; $\beta = 0.004$, $p = 0.990$ continuous). While not statistically significant, the direction is consistent across specifications. The entry effect remains stable at 16.7 pp across Tier 1 and Tier 2, suggesting contact patterns do not fully mediate the entry effect. Reverse causality cannot be ruled out.

Tier 3 — Exploratory (H3 + H4, n = 393–395, already-transnational only). Distance alone was not significantly associated with loneliness change ($\beta = 0.009$, $p = 0.726$). However,

the satisfaction $\times$ distance interaction was **highly significant** ($\beta = -0.464$, $p < 0.001$). Among households where satisfaction deteriorated, each additional 1,000 km increased loneliness by 0.481 points ($p < 0.001$). Among households where satisfaction improved, greater distance was associated with *lower* loneliness (-0.448 per 1,000 km, $p < 0.001$). These findings are preliminary given the small and structurally selected subsample.

15.2 Limitations

- **Time-varying unobserved confounding (Tier 1):** Simultaneous health deterioration or economic shock could cause both migration and increasing loneliness, and cannot be ruled out without an instrument. IV using POEA municipality-level deployment data is a planned extension.
- **Reverse causality (Tier 2):** Loneliness may reduce contact satisfaction rather than satisfaction buffering loneliness. Panel mediation with lagged variables would address this with three or more waves.
- **Structural missingness (Tier 3):** 92% of the panel is missing destination country codes in at least one wave. The distance subsample ($n = 395$) is not representative of broader transnational households and contains zero newly-entering households, limiting generalisation.
- **Statistical power:** With 223 enterers in the complete-case Tier 1 sample, the study is powered to detect standardised effects of approximately 0.20. Smaller true effects would require a larger panel or additional waves.
- **Attrition:** Wave 2 has substantially more missing values on the 3-item loneliness scale ($NA = 2,567$ vs 775 at Wave 1), likely reflecting proxy interviews for frailer respondents. If attrition is non-random with respect to loneliness, Tier 1–2 estimates may understate the loneliness burden in the most vulnerable subgroup.

16 Session Information

```
sessionInfo()
```

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Matrix products: default
LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_Netherlands.utf8 LC_CTYPE=English_Netherlands.utf8
[3] LC_MONETARY=English_Netherlands.utf8 LC_NUMERIC=C
```

[5] LC_TIME=English_Netherlands.utf8

time zone: Asia/Manila
tzcode source: internal

attached base packages:

[1] grid stats graphics grDevices utils datasets methods
[8] base

other attached packages:

[1] patchwork_1.3.2	kableExtra_1.4.0	knitr_1.51
[4] writexl_1.5.4	rsvg_2.7.0	DiagrammeRsvg_0.1
[7] DiagrammeR_1.0.11	sf_1.0-23	rnaturalearth_1.1.0
[10] geosphere_1.5-20	modelsummary_2.6.0	marginaleffects_0.32.0
[13] cobalt_4.6.2	WeightIt_1.6.0	survey_4.4-8
[16] survival_3.8-3	Matrix_1.7-4	broom_1.0.11
[19] forcats_1.0.1	haven_2.5.5	janitor_2.2.1
[22] scales_1.4.0	ggplot2_4.0.1	tidyr_1.3.2
[25] dplyr_1.1.4		

loaded via a namespace (and not attached):

[1] DBI_1.2.3	sandwich_3.1-1	rlang_1.1.6
[4] magrittr_2.0.4	snakecase_0.11.1	otel_0.2.0
[7] e1071_1.7-17	compiler_4.5.2	systemfonts_1.3.1
[10] vctrs_0.6.5	stringr_1.6.0	pkgconfig_2.0.3
[13] fastmap_1.2.0	backports_1.5.0	labeling_0.4.3
[16] utf8_1.2.6	rmarkdown_2.30	tzdb_0.5.0
[19] ragg_1.5.0	purrr_1.2.0	xfun_0.55
[22] jsonlite_2.0.0	tinytable_0.16.0	chk_0.10.0
[25] R6_2.6.1	tables_0.9.33	stringi_1.8.7
[28] RColorBrewer_1.1-3	lmtest_0.9-40	lubridate_1.9.4
[31] estimability_1.5.1	Rcpp_1.1.0	zoo_1.8-15
[34] parameters_0.28.3	readr_2.1.6	splines_4.5.2
[37] timechange_0.3.0	tidyselect_1.2.1	rstudioapi_0.17.1
[40] yaml_2.3.12	codetools_0.2-20	curl_7.0.0
[43] lattice_0.22-7	tibble_3.3.0	withr_3.0.2
[46] bayestestR_0.17.0	S7_0.2.1	evaluate_1.0.5
[49] rnaturalearthdata_1.0.0	units_1.0-0	proxy_0.4-29
[52] xml2_1.5.1	pillar_1.11.1	KernSmooth_2.23-26
[55] checkmate_2.3.4	insight_1.4.6	generics_0.1.4
[58] sp_2.2-0	hms_1.1.4	xtable_1.8-8
[61] class_7.3-23	glue_1.8.0	emmeans_2.0.2
[64] tools_4.5.2	data.table_1.18.0	mvtnorm_1.3-3
[67] visNetwork_2.1.4	mitools_2.4	datawizard_1.3.0
[70] performance_0.16.0	cli_3.6.5	textshaping_1.0.4
[73] viridisLite_0.4.2	svglite_2.2.2	V8_8.0.1
[76] gtable_0.3.6	digest_0.6.39	classInt_0.4-11
[79] htmlwidgets_1.6.4	farver_2.1.2	htmltools_0.5.9

[82] lifecycle_1.0.5